INCH-POUND

MIL-P-70343
22 March 1990

MILITARY SPECIFICATION

PACKING, FIBERGLASS, ROPE AND WICK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements and tests for rope and wick fiberglass packing used for boiler seals for service temperatures up to 1000°F.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping-Steel, and Seals
PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-601 - Boxes, Wood, Cleated, Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-BAC-S, Picatinny Arsenal, NJ 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5330

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MILITARY

MIL-P-116 - Preservation - Packaging, Methods of

STANDARDS

FEDERAL

FED-STD-102 - Preservation, Packaging and Packing Levels

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-794 - Part and Equipment, Procedures for Packaging and Packing of

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ASTM

ASTM D 578 - Standard Specifications for Glass Fiber Yarns
ASTM D 3951 - Practice for Commercial Packing
ASTM D 4268 - Standard Method of Testing Fiber Rope

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. The material shall be continuous, high density, unsintered, textured (T) or carded yarns made from Type STDE fiberglass material. Asbestos and components containing asbestos are prohibited.

3.2 Construction. The packing shall be composed of twisted plied strands, laid up into the general form of a wick or a loosely twisted rope according to the manufacturer's design. If a core is employed, it shall be of the same material and construction.

3.3 Size and weight. The size and weight shall be as shown in Table I.

TABLE I. SIZE AND WEIGHT

DASH NO	SIZE DIAMETER (INCHES) ±10%		FT/LB ±10%	UNIT WEIGHTS
	FRACTION	DECIMAL		
	Wick	Wick		13 OR 25 Pound
001W	1/4	.250	54.0	Spool, Reel, or Coil
002W	3/8	.375	29.0	
	Rope	Rope		
001R	1/4	.250	55.0	
002R	3/8	.375	23.5	
003R	1/2	.500	14.0	
004R	5/8	.625	9.0	25 or 50 Pound
005R	3/4	.750	6.5	Spool, Reel, or Coil
006R	7/8	.875	4.5	
007R	1	1.000	3.5	
008R	1-1/4	1.250	2.5	
009R	1-1/2	1.500	2.0	
010R	2	2.000	1.2	

3.4 Workmanship. The workmanship shall be such that the packing shall be free from defects which may affect its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2. Quality conformance inspection. Quality conformance inspections shall be as specified in Table II.

TABLE II. QUALITY CONFORMANCE INSPECTION

	INSPECTION	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
GROUP A	DIMENSIONS.... VISUAL....	TABLE I 3.4	4.3.2 4.4
GROUP C	MATERIAL....	3.1	4.3.1

4.2.1 Lot. For purpose of sampling, a lot shall consist of all packing under the same size produced under essentially the same conditions and offered for delivery at one time.

4.2.2 Sampling for visual, dimensional and weight examination. A random sample of spools, reels or coils shall be selected from each lot offered for inspection with lot acceptance based on inspection requirements in accordance with MIL-STD-105, Inspection Level II. Any spool, reel or coil containing one or more visual, dimensional or weight defects shall not be offered for delivery. This shall be cause for rejection of the lot represented by the sample.

4.2.3 Sampling for tests. A random sample of spools, reels or coils shall be selected from each lot in accordance with Table III for the test specified in 4.3. If any specimen tested is found not to be in conformance with the requirements specified in the specification, this shall be cause for rejection of the lot represented by the sample.

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TABLE III. SAMPLING FOR TESTS.

NUMBER OF SPOOLS, REELS OR COILS IN LOT	NUMBER OF SPOOLS, REELS OR COILS IN SAMPLE 1/
100 and under	3
101 to 300	5
301 to 500	7
501 and over	10

1/ From each sample spool, reel or coil, a specimen weighing not less than 6 ounces shall be taken.

4.3 Test methods.

4.3.1 Material. Material shall be tested in accordance with ASTM D 578.

4.3.2 Size. Material shall be measured using precision measuring tools, such as a micrometer and shall be in accordance with the size specified in Table I.

4.3.3 Splices. Unless otherwise agreed upon there shall be allowed a maximum of 2 splices per unit with no piece shorter than 10 feet in length.

4.4 Visual inspection. The packing shall be visually inspected to assure that it is free of all defects such as cuts, chafes, marks, stains or spots that may affect serviceability. Packing material shall also be free of any embedded foreign matter.

4.5 Packaging inspection. An inspection shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with Table IV. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II and a single defect shall be cause for the rejection of the entire lot.

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TABLE IV. CLASSIFICATION OF PREPARATION FOR DELIVERY DEFECTS

INSPECTION	DEFECT
Marking (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, or inadequate stapling. Bulging or distortion of container.

5. PACKAGING

5.1. Packaging requirements. For civil agency procurement the definitions and application of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.2 Preservation. Preservation shall be level A, C, or Commercial as specified (see 6.2).

5.2.1 Level A. Packing of like description furnished on spools, reels or coils shall be packaged individually in a close fitting box in accordance with PPP-B-636, Class 2, Style RSC. The box shall be closed in accordance with the box specification.

5.2.1.1 Cleaning. Spools, reels or coils shall be cleaned in accordance with MIL-P-116, process C-1.

5.2.1.2 Drying. Spools, reels or coils shall be dried in accordance with MIL-P-116.

5.2.1.3 Unit packs. Unless otherwise specified (see 6.2), each packing of like description furnished on spools, reels, or coils shall be individually unit packed in accordance with Method III of MIL-P-116 designated herein insuring compliance with the applicable requirements of that specification.

5.2.2 Level C. The level C preservation for the spools, reels or coils shall conform to the MIL-STD-794 requirements for this level.

5.2.3 Commercial. The commercial preservation of spools, reels or coils shall be in accordance with the requirements of ASTM D 3951.

5.3 Packing. Packing shall be level A, B, C or Commercial, as specified (see 6.2).

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5.3.1 Level A. The packaged spools, reels or coils shall be packed in containers conforming to PPP-B-601, overseas type; PPP-B-621, Class 2, Style 4 or PPP-B-585, Class 3, Style 2 or 3. Closure and strapping shall be in accordance with applicable container specifications except the strapping shall conform to QQ-S-781, Type I or IV, Finish A. When the Gross Weight exceeds 200 pounds or the container length and width is 48" X 24" or more and the weight exceeds 100 pounds, 3 X 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification.

5.3.2 Level B. The packaged spools, reels or coils shall be packed in fiberboard containers conforming to PPP-B-636, Class weather resistant, Style optional, Special requirements; or PPP-B-601 Domestic type; PPP-B-621, domestic; PPP-B-585, Class domestic. Closure and strapping shall be in accordance with applicable container specifications except the strapping shall conform to QQ-S-781, Type I or IV, Finish A. In lieu of the closure and waterproofing requirements in the Appendix of PPP-B-636, closure waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joints with Waterproof Packaging Tape, or Box Closure and Sealing Tape, two inches minimum width. When gross weight exceeds 200 pounds or the container length and width is 48" X 24" or more and the weight exceeds 100 pounds, 3 X 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification.

5.3.3 Level C. The level C packing for spools, reels or coils shall conform to MIL-STD-794 requirements for this level.

5.3.4. Commercial. The preserved spools, reels or coils shall be packed in accordance with requirements of ASTM D 3951.

5.3.5 Palletized loads. When specified (see 6.2) palletized loads, commensurate with the level of packing specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking.

5.4.1 Level A, B, and C. In addition to any special or other identification marking required by the contract (see 6.2) each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4.2 Commercial. Commercial markings shall be in accordance with the requirements of ASTM D 3951.

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5.5 General.

5.5.1 Exterior containers. Exterior containers (see 5.3.1, 5.3.2, 5.3.3, and 5.3.4) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possible.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers packing for use as a sealer between steam sectional boiler heads, uptake dampers, grommets for boiler casing bolting, and in the construction for boiler casing gaskets, for service temperatures up to 1000°F.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Size and weight of coil required (see 3.3).
c. Selection of applicable levels of preservation, packaging, packing, marking and commercial/industrial (see 5.2, 5.3, and 5.4).
d. When palletized loads are required (see 5.3.5).

6.3 Part or Identifying Number (PIN). The PIN to be used for packing acquired to this specification is created as follows:

M	70343	001W
Prefix to indicated military specification	Specification number	Dash number from Table I

Part number example:

M70343-001W indicates .250 (1/4 inch) diameter wick fiberglass packing, on a 13 or 25 pound spool, reel or coil.

6.4 Subject term (key word) listing.

Yarn
Unsintered
Braided
Plied
Strands

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70343	22 March 1990

3. DOCUMENT TITLE

PACKING, FIBERGLASS, ROPE AND WICK

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

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